TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the future: An exploration wise consumption pattern
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Performance Testing

Performance testing was conducted to evaluate the efficiency, responsiveness, and reliability of the Electricity Consumption Dashboard. The testing focused on three major aspects: the amount of data rendered from the database, the utilization of data filters, and the number of visualizations displayed. Optimizing these components ensures smooth dashboard performance and timely delivery of insights.

Amount of Data Rendered to DB

The volume of data retrieved and rendered from the database was monitored to ensure efficient query execution and system performance. The amount of data rendered depends on the dataset size and the database's capability to store and retrieve information. Using MySQL Workbench, database details such as table rows, column count, and storage information were analysed by selecting the required table and viewing its properties. This helped verify that the database could efficiently handle the project dataset.

Utilization of Data Filters

Data filters were implemented in Tableau to enhance dashboard interactivity and improve performance. Filters such as State, Region, Year, and Date allow users to focus on specific portions of the dataset, reducing the amount of data processed at a time. Additional Top N and Bottom N filters were used to identify the highest and lowest electricity-consuming states. These filters improve responsiveness while enabling users to perform detailed and meaningful analysis.

Number of Visualizations / Graphs

The dashboard contains multiple visualizations designed to analyse electricity consumption patterns from different perspectives. The visualizations developed include:

- 2019 State Consumption
- 2020 State Consumption
- Total Consumption
- Usage by Region
- Top N and Bottom N Analysis
- 2019 and 2020 Month-wise Consumption
- Total Consumption Region-wise
- Usage Before and After Lockdown
- Region-wise State Usage
- Quarter-wise Usage
- Metro City State Usage
- Usage by Year

These visualizations were optimized to ensure smooth dashboard performance while providing comprehensive insights into electricity consumption trends across different regions and time periods.